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ORIGINAL ARTICLE – MELANOMA

Immune Status in Merkel Cell Carcinoma: Relationships With Clinical Factors and Independent Prognostic Value

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ABSTRACT

Background. Immunosuppression (IS) currently is not considered in staging for Merkel cell carcinoma (MCC). An analysis of the National Cancer Database (NCDB) was performed to investigate immune status as an independent predictor of overall survival (OS) for patients with MCC and to describe the relationship between immune status and other prognostic factors.

Methods. The NCDB was queried for patients with a diagnosis of MCC from 2010 to 2016 who had known immune status. Multivariable Cox proportional hazards models were used to define factors associated with OS. Secondary models were constructed to assess the association between IS etiology and OS. Multivariable logistic regression models were used to characterize relationships between immune status and other factors.

Results. The 3-year OS was lower for the patients with IS (44.6%) than for the immunocompetent (IC) patients (68.7%; $p < 0.0001$). Immunosuppression was associated with increased adjusted mortality hazard (hazard ratio [HR], 2.36, 95% confidence interval [CI], 2.03–2.75). The etiology of IS was associated with OS ($p = 0.0015$), and patients with solid-organ transplantation had the lowest 3-year OS (32.7%). Immunosuppression was associated with increased odds of greater nodal burden (odds ratio [OR], 1.70; 95% CI, 1.37–2.11) and lymphovascular invasion (OR, 1.58; 95% CI, 1.23–2.03).

Conclusions. Immune status was independently prognostic for the OS of patients with localized MCC. The etiology of IS may be associated with differential survival outcomes. Multiple adverse prognostic factors were associated with increased likelihood of IS. Immune status, and potentially the etiology of IS, may be useful prognostic factors to consider for future MCC staging systems.

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Merkel cell carcinoma is a rare cutaneous neuroendocrine malignancy with increasing incidence.[1] Immunosuppression (IS) is related to increased risk of MCC development,[2,3] with the host immune system believed to be important in MCC pathogenesis.[4] Immunosuppression also has been related to worsened clinical outcomes in multiple published studies,[5–7] with national guidelines from the National Comprehensive Cancer Network (NCCN) and the American Joint Committee on Cancer (AJCC) recommending consideration of immune status in the development of treatment and surveillance plans.[8] However, immune status is not currently incorporated into AJCC staging for MCC patients, with AJCC authors citing the subjective nature of immunosuppression as a key limitation.[9] In addition, inclusion of immune status in staging may be complicated by limited characterization of immunosuppressive etiology and potential differential risk of clinical detriment for patients with MCC.

Given the relative rarity of MCC, many prior investigations have been limited in ability to characterize associations between immunosuppressive etiology and clinical outcomes.[5–7,10] Furthermore, the relationship between immune status and other known clinical prognostic factors such as positive lymph nodes, tumor size, and lymphovascular invasion have not been fully described for patients with MCC.

We performed an analysis of the National Cancer Database (NCDB)[11] to define predictors of overall survival (OS) including immune status for patients with localized MCC (stages I–III) to better characterize the association between immunosuppressive etiology and OS and to further describe the relationships between immune status and other known clinical factors for these patients. Increased understanding of the aforementioned relationships may have high clinical relevance with regard to improved prognostication for patients with localized MCC and may offer additional support for considering incorporation of immune status in formal staging for these patients. An improved definition of immune status as an independent prognostic factor in MCC also may help to guide patient cohort stratification for future trials investigating optimal treatment selection and surveillance strategies. This may be of particular relevance given the relative contraindications to receipt of immunotherapy for select groups of IS patients with MCC such as those with solid-organ transplants.

MATERIALS AND METHODS

The NCDB was queried for patients with localized MCC (stages I, II, or III) diagnosed from 2010 to 2016 who underwent surgical resection with known immune status and details regarding radiotherapy and receipt of chemotherapy (selected factors depicted in Table 1, full models available as Supplementary material). Immune status was obtained from the dataset using collaborative stage (CS) site-specific factors coding for etiologies underlying profound immunosuppression. The study used CS Site-Specific Factor 22, which codes for no immune suppression condition or conditions, human immunodeficiency virus (HIV)/acquired immunodeficiency syndrome (AIDS), solid-organ transplant recipient, chronic lymphocytic leukemia, non-Hodgkin lymphoma, more than one of the aforementioned conditions, other specified diagnosis resulting in profound immunosuppression, profound immune suppression present with the diagnosis not recorded, and unknown/no information collected.

Profound immunosuppression is coded with regard to AJCC recommendations, and coders are instructed to review physician consultation notes, patient histories, and additional documents available in the electronic medical record to determine whether a patient is listed as immunocompromised, immunosuppressed, or experiencing immunosuppression. In addition, these conditions must be active or must have been diagnosed within 2 years after MCC diagnosis to be coded for profound immunosuppression.[12]

Similarly, tumor-infiltrating lymphocyte (TIL) grade was abstracted from the dataset via collaborative stage site-specific factors coding for tumor specimen histopathologic examination (categorized as absent, non-brisk, or brisk). The TIL grade is a semi-quantitative factor recommended by the AJCC to be documented in histopathologic reports.[9] Absence of the TIL grade was defined as no lymphocyte-admixed tumor cells present in the specimen (even if lymphocytes were present in perivascular areas within the tumor or beyond the tumor border but not abutting tumor cells). Brisk TIL grade was defined as the presence of infiltrating lymphocytes along 90% of the circumference of the lesion base or the presence of a diffuse lymphocyte infiltrate throughout the vertical growth phase of the lesion. Specimens with lymphocyte infiltrates not meeting the aforementioned criteria for brisk TIL grade were categorized as non-brisk TIL grade.[9] Coders are instructed to abstract the TIL grade from histopathology reports in the electronic medical record in accordance with the aforementioned guidelines.

Surgical margin status after definitive resection was abstracted from the dataset, with patients listed in the study cohort as having microscopic residual tumor, residual tumor not otherwise specified (NOS), or macroscopic residual tumor categorized as a tumor remnant remaining after definitive surgical resection. The study excluded patients receiving neoadjuvant chemotherapy or radiation therapy (Fig. S1). Patients who received adjuvant radiation therapy (RT) doses larger than 80 Gy or less than 30 Gy also were excluded to limit the potential inclusion of patients receiving atypical doses of adjuvant RT.

The study abstracted and adjusted for patient demographic, tumor, and treatment factors in the performed analyses. Descriptive statistics were calculated to characterize differences regarding the aforementioned clinical factors present in the study cohort stratified by immune status.

Overall survival was defined as the time from resection to death from any cause or censoring. Kaplan-Meier methods and the log-rank test were performed to compare the OS for the study cohort stratified by immune status. A multivariable Cox proportional hazards regression model was fitted to identify factors related to OS, with adjustment

TABLE 1 Baseline patient, tumor, and treatment factors[a]

	Total		Immunocompetent		Immunosuppressed		*p* value
	n	Prop (%)	*n*	Prop (%)	*N*	Prop (%)	
	3882		3470	89.4	412	10.6	
Radiation treatment							**0.0261**
No	1864	48.0	1688	48.6	176	42.7	
Yes	2018	52.0	1782	51.4	236	57.3	
Chemotherapy							0.2908
No	3658	94.2	3275	94.4	383	93.0	
Yes	224	5.8	195	5.6	29	7.0	
Age (years)			Median: 75		Median: 72		**0.0003**
< 60	396	10.2	344	9.9	52	12.6	
60–69	898	23.1	787	22.7	111	26.9	
70–79	1265	32.6	1118	32.2	147	35.7	
80+	1323	34.1	1221	35.2	102	24.8	
Sex							**0.0019**
Male	2428	62.5	2141	61.7	287	69.7	
Female	1454	37.5	1329	38.3	125	30.3	
Race							0.1872
White	3775	97.2	3379	97.4	396	96.1	
Non-white	107	2.8	91	2.6	16	3.9	
Charlson/Deyo comorbidity score							**0.0028**
0	2738	70.5	2472	71.2	266	64.6	
1	787	20.3	696	20.1	91	22.1	
2+	357	9.2	302	8.7	55	13.3	
Stage							**0.0068**
1	1986	51.2	1800	51.9	186	45.1	
2	720	18.5	646	18.6	74	18.0	
3	1176	30.3	1024	29.5	152	36.9	
Surgical margins							0.7848
No residual tumor	3451	90.4	90.4			89.8	
Residual tumor	365	9.6	9.6			10.2	
Unknown	66						
Tumor size (cm)							0.1113
< 1	977	33.3		34.6		29.8	
1.0–1.9	974	33.2		33.5		35.4	
2.0–3.9	712	24.3		23.8		23.4	
4+	267	9.1		8.2		11.5	
Unknown	952						
Subsite							**0.0326**
Head/neck	1679	43.3	1518	43.7	161	39.1	
Trunk	375	9.7	322	9.3	53	12.9	
Limbs/other	1828	47.1	1630	47.0	198	48.1	
TIL							**0.0243**
Absent	1206	71.2		57.1		51.3	
Non-brisk	353	20.8		30.8		41.6	
Brisk	135	8.0		12.1		7.2	
Unknown	2188						
Nodes							**< 0.0001**
None positive	1797	62.7	54.8			45.7	

TABLE 1 continued[a]

	Total		Immunocompetent		Immunosuppressed		*p* value
	n	Prop (%)	*n*	Prop (%)	*N*	Prop (%)	
1–3 positive	871	30.4	31.4			31.0	
4–8 positive	128	4.5	7.7			11.6	
≥9 positive	68	2.4	6.1			11.7	
Unknown	1018						
ECS							**0.0002**
No ECS	653	23.5	24.4			34.5	
ECS	325	11.7	20.9			19.7	
No positive nodes	1797	64.8	54.8			45.7	
Unknown	1107						
LVI							**0.0007**
Not present	1771	63.9	65.4			55.2	
Present	999	36.1	34.6			44.8	
Unknown	1112						
Radiation treatment to lymph nodes							**0.0397**
RT without nodal coverage	1182	33.0	35.3			41.4	
Nodal RT	537	15.0	16.1			15.9	
No radiation	1864	52.0	48.6			42.7	
Unknown	299						
Immunosuppression type							
None	3470	89.4	3470	100.0	—	0.0	
Organ transplant	106	2.7	—	0.0	106	25.7	
Leukemia	118	3.0	—	0.0	118	28.6	
Non-Hodgkins Lymphoma	72	1.9	—	0.0	72	17.5	
Other/NOS/multiple	116	3.0	—	0.0	116	28.2	

Statistically significant *p* values are in bold

Prop, proportion; residual tumor, final surgical margin status after definitive surgical resection as residual tumor NOS, microscopic residual tumor, or macroscopic residual tumor; TIL, tumor-infiltrating lymphocyte; ECE, extracapsular extension; LVI, lymphovascular invasion; RT, radiation therapy; NOS, not otherwise specified; ECS, extracapsular spread; Other/NOS/multiple, immunosuppression (IS) group included CS Site-Specific Factor 22 coding for HIV/AIDS, more than one of the aforementioned conditions, other specified diagnosis resulting in profound immunosuppression, profound immune suppression present with diagnosis not recorded

[a]Variables with unknown status were imputed using multiple imputation. The percentage values stratified by TIL grade are based on the imputed datasets and thought to better represent population-averaged proportions for these factors

for a priori chosen factors including immune status, adjuvant radiation and chemotherapy, age, sex, race, final surgical margin status, pathologic tumor size, anatomic subsite, TIL grade, number of positive lymph nodes, extracapsular extension (ECE), and lymph-vascular invasion (LVI) (Table 2). These adjusted factors from the mulivariable Cox proportional hazards regression model were each delineated as a hazard ratio (HR). Adjuvant radiation and chemotherapy were treated as time-dependent predictors to limit immortal time bias.[13,14] Schoenfeld residuals were evaluated to ensure non-violation of the proportional hazards assumption.

The final multivariable Cox regression model was stratified by stage, after which the model satisfied the global proportional hazard test.[15] Interaction effect testing was performed to determine whether differences were present in the study cohort with regard to the effect of immune status and OS by other predictors. Interaction effect testing also was used to determine whether a differential effect was present in the cohort regarding etiology of immunosuppression and OS by substitution of the immune status variable in the full Cox regression with a separate effect for each etiology of immunosuppression.

TABLE 2 Multivariable Cox proportional hazards regression model for selected factors associated with overall survival[a]

	Adj HR	95% CI			*p* value
Immune status				< **0.0001**	
Immunocompetent	Reference				
Immunosuppressed	2.36	2.03	2.75		< **0.0001**
Radiation treatment				< **0.0001**	
No	Reference				
Yes	0.73	0.65	0.83		< **0.0001**
Chemotherapy				0.0582	
No	Reference				
Yes	1.25	0.99	1.57		0.0582
Age (years)				< **0.0001**	
< 60	Reference				
60–69	1.46	1.10	1.93		**0.0096**
70–79	1.98	1.48	2.65		< **0.0001**
80+	4.10	3.06	5.49		< **0.0001**
Sex				< **0.0001**	
Male	Reference				
Female	0.73	0.65	0.82		< **0.0001**
Charlson/Deyo comorbidity score				< **0.0001**	
0	Reference				
1	1.22	1.08	1.39		**0.0022**
2+	1.69	1.41	2.02		< **0.0001**
Surgical margins				< **0.0001**	
No residual tumor	Reference				
Residual tumor	1.45	1.21	1.74		< **0.0001**
Subsite				< **0.0001**	
Head/neck	Reference				
Trunk	1.09	0.91	1.30		0.3622
Limbs/other	0.77	0.68	0.86		< **0.0001**
Nodes				< **0.0001**	
None positive	Reference				
1–3 positive	1.34	1.05	1.71		**0.0200**
4–8 positive	1.82	1.41	2.35		< **0.0001**
≥9 positive	2.51	1.81	3.49		< **0.0001**
Extracapsular extension				**0.0003**	
No ECS	Reference				
ECS	1.39	1.16	1.65		**0.0003**
Vascular invasion				**0.0116**	
Not present	Reference				
Present	1.18	1.04	1.34		**0.0116**
Stage					
Stage effect violates proportional hazards assumption ($p < 0.0001$).					
Effect is included as a strata variable with stage-dependent baseline hazards functions.					

Statistically significant *p* values are in bold

Adj HR, adjusted hazard ratio; CI, confidence interval; ECS, extracapsular extension; residual tumor, final surgical margin status after definitive surgical resection as residual tumor NOS, microscopic residual tumor, or macroscopic residual tumor

[a]This Cox model also controlled for effects of race, diagnosis year, tumor size, and tumor-infiltrating lymphocyte. These effects were found to be nonsignificant and suppressed from this table (see Supplementary material for full model results). The model also included a strata effect for stage because it violates the proportional hazards assumption ($p < 0.0001$)

TABLE 3 Multivariable regression models of relationships between immune status and other clinical factors

	Adjusted OR due to IS			
	AOR	LCI	UCI	p value
IS association with clinical factors				
Vascular invasion[a]	1.58	1.23	2.03	**0.0004**
Extracapsular extension[a,b]	0.70	0.50	0.99	**0.0439**
Number of nodes[a,c]				
Proportional OR	1.70	1.37	2.11	< **0.0001**
Tumor size[a,c]				
Proportional OR	1.15	0.93	1.42	0.2116
TIL[a]				
Non-brisk vs absent	1.54	0.76	3.09	0.2281
Brisk versus absent	0.61	0.21	1.80	0.3742
IS association with treatment received				
Radiation treatment[d]	1.10	0.87	1.38	0.4232
Chemotherapy[d]	0.70	0.47	1.24	0.2783
Nodal radiation[d,e]	0.71	0.48	1.03	0.0723

Statistically significant p values are in bold

OR, odds ratio; IS, immunosuppression; AOR, adjusted OR; LCI, lower confidence interval; UCI, upper confidence interval; TIL, tumor-infiltrating lymphocyte

[a]Model controls for age, sex, race, Charlson score, and subsite

[b]Only patients with positive nodes were included

[c]Proportional odds model used for ordinal response variable. The proportional odds ratio represents the increased odds of moving to a higher/worse category due to IS

[d]Model controls for insurance type, diagnosis year, age, sex, race, Charlson score, subsite, vascular invasion, extracapsular extension, number of nodes, tumor size, and TIL

[e]Only patients who received RT were included

Multivariable regression models were used to assess relationships between immune status and the likelihood of other known clinically relevant factors (Table 3). Models used to assess relationships between immune status and clinical factors (LVI, ECS, lymph node numbers, tumor size, and TIL grade) were adjusted for the a priori selected covariates (age, sex, race, Charlson/Deyo comorbidity score, and anatomic subsite). Models used to assess relationships between immune status and the likelihood of treatment also were adjusted for insurance type, diagnosis year, LVI, ECS, positive lymph node number, tumor size, and TIL grade in addition to the aforementioned factors. Logistic regression was used for binary variables (LVI, ECS, RT, chemotherapy, nodal RT). Proportional odds logistic regression was used for ordinal responses (node and tumor size), and baseline category logit models were used for unordered categorical variables (TIL grade).

Throughout all the analyses, multiple imputation with a chained-equations approach[16] was performed to account for missing data bias using the mice and miceadds packages for R.[17,18] Statistical analyses were performed using the R statistical software v3.6.2.[19] The tests were two-sided. Statistical significance was defined by an alpha of 0.05 or lower. This study was approved by the institutional review board (IRB) and conducted in accordance with Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.[20]

RESULTS

Study Cohort

Due to missing or unknown immune status, 18,482 patients were excluded from the pre-selection criteria cohort. The study cohort included 3882 patients (3470 patients with known immunocompetence and 412 patients with known immunosuppression; Table 1). Regarding lymph node examination, 2864 patients (73.8%) underwent surgical nodal examination. The median follow-up time was 33 months (interquartile range [IQR], 18–55 months). The etiologies for profound immunosuppression included chronic lymphocytic leukemia (CLL, n = 118), HIV/AIDS (n = 116), solid-organ transplant (n = 106), and non-Hodgkin lymphoma (NHL, n = 72). The 3-year OS rate for the overall cohort was 66.2% (95% confidence interval [CI], 64.6–67.8%), with the stage 1 patients having a higher median 3-year OS rate (73.9%; 95% CI, 71.9–76.0%) than the stage 2 patients (61.8%; 95% CI, 58.2–-66.6%) or stage 3 patients (55.9%, 95% CI, 53–58.9% ($p < 0.0001$; Fig. S2). Adjuvant RT was received by 52% of the overall cohort (n = 2018). The majority of the patients in the cohort receiving RT did not receive RT that targeted draining lymph nodes (n = 1182), with 573 patients receiving RT to draining lymph nodes and 299 patients having unknown status regarding RT that targeted draining lymph nodes.

Immune Status and Overall Survival

The patients with IS had significantly lower 3-year OS rates (44.6%; 95% CI, 39.8–49.9%) than the immunocompetent patients (68.7%; 95% CI, 67.1–70.4%; $p < 0.0001$; Fig. 1). Immunosuppression was associated with an increased adjusted risk of mortality (hazard ratio [HR], 2.36; 95% CI, 2.03–2.75) relative to immunocompetence (reference group). The etiology of immunosuppression was associated with OS in a secondary Kaplan-Meier analysis (p = 0.0015; Fig. 2A), with lower 3-year OS rates for the IS

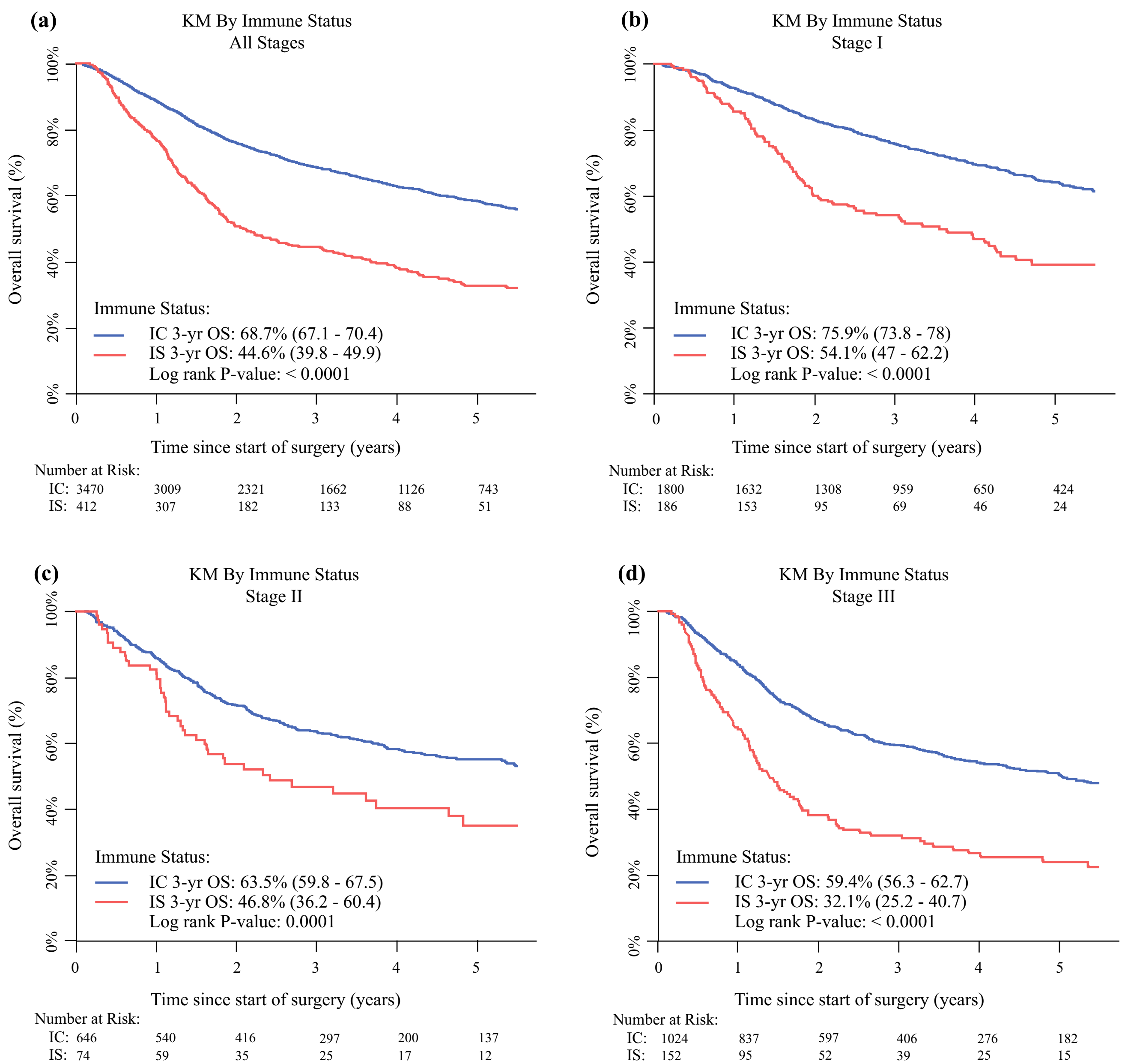

FIG. 1 **a** Kaplan-Meier curve of overall survival (OS) depicted for the overall cohort stratified by immune status. **b** Kaplan-Meier curve of OS depicted for stage 1 patients stratified by immune status. **c** Kaplan-Meier curve of OS depicted for stage 2 patients by immune status. **d** Kaplan-Meier curve of OS depicted for stage 2 patients by immune status. IS, immune suppressed; IC: immunocompetent

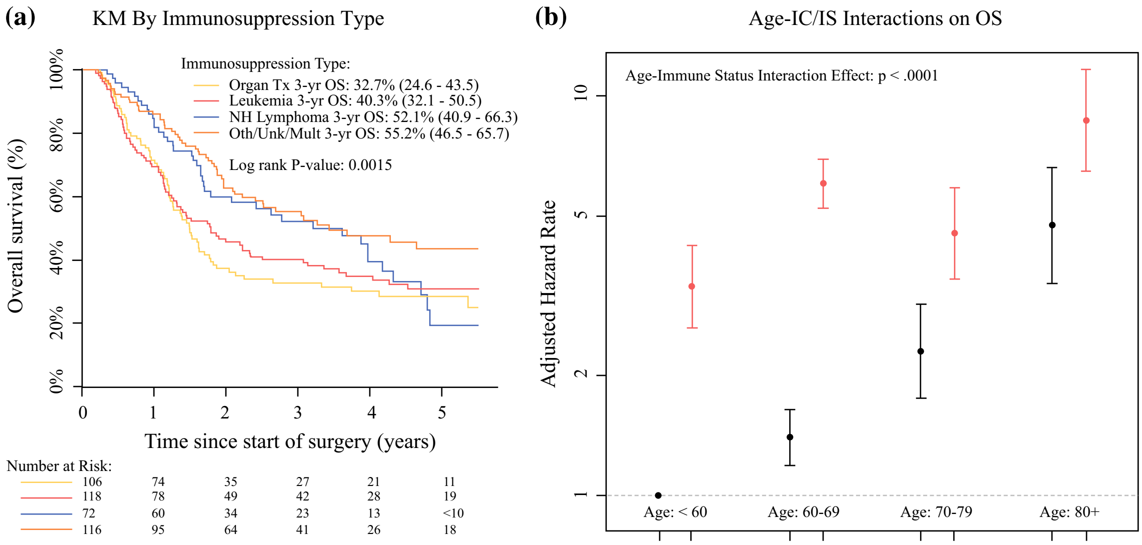

FIG. 2 a Kaplan-Meier curve of overall survival depicted for immune-suppressed patients stratified by etiology of immunosuppression. Mult, multiple; NH, non-Hodgkin's lymphoma; Oth, other; OS, overall survival; Tx, treatment; Unk, unknown; yr, year. **b** Interaction testing effect of age and immune status on overall survival. IS, immune suppressed; IC, immunocompetent

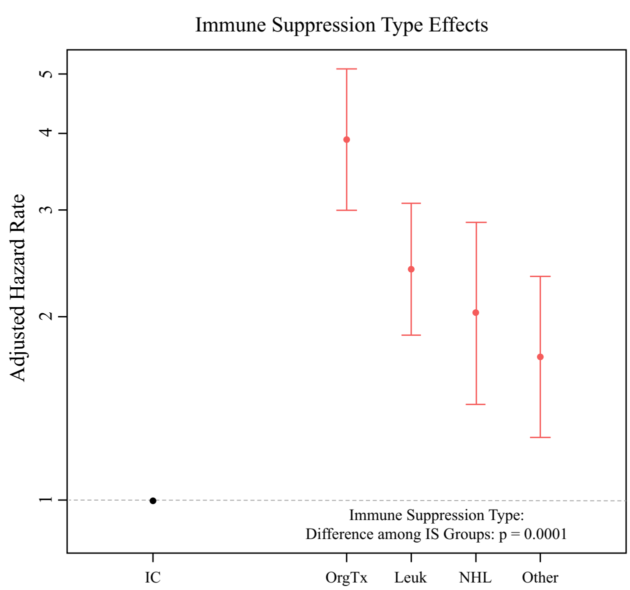

FIG. 3 Interaction testing effect that the etiology of immunosuppression has on overall survival. IS, immune suppressed

patients with a solid-organ transplant (32.7%; 95% CI, 24.6–43.5%) than the patients with CLL (40.3%; 95% CI, 32.1–50.5%), NHL (52.1%; 95% CI, 40.9–66.3%), or HIV/AIDS/other (55.2%; 95% CI, 46.5–65.7%). Adjusted effect-testing suggested that solid-organ transplantation may have increased the adjusted mortality hazard relative to other immunosuppressive etiologies ($p = 0.0001$; Fig. 3) when control was used for other clinical factors. Interaction effect-testing also suggested a differential effect of immune status on OS by patient age, with worsened mortality for immunosuppressed (IS) patients relative to immunocompetent (IC) patients younger than 60 years and those 60 to 69 years old than for patients 70 to 79 years old and those 80 years old or older ($p < 0.0001$; Fig. 2B). Significant interaction effects with IS were not found for any other predictors.

Additional Clinical Factors and Overall Survival

Adjuvant RT, age, sex, Charlson/Deyo comorbidity score, surgical margin status, anatomic subsite, positive lymph nodes, ECS, and LVI all were significantly associated with OS by multivariable Cox regression modeling (all p values < 0.05). Adjuvant RT was associated with decreased adjusted mortality hazard (HR, 0.73; 95% CI, 0.65–0.83). Residual tumor (HR, 1.45; 95% CI, 1.21–1.74), ECS (HR, 1.39; 95% CI, 1.16–1.65), LVI (HR, 1.18; 95% CI, 1.04–1.34), and positive lymph nodes (1 to 3 positive nodes: HR, 1.34; 95% CI, 1.05–1.71; 4 to 8 nodes: HR, 1.82; 95% CI, 1.41—2.35; 9 or more nodes: HR, 2.51; 95% CI, 1.81–3.49) all were associated with increased adjusted mortality hazard compared respectively with no residual tumor, no ECS, no LVI, and no involved lymph nodes. Extremities and other anatomic subsites were associated with decreased mortality hazard with respect to head and neck MCC (HR, 0.77; 95% CI, 0.68-0–86). Each decade of age was associated with increased mortality hazard with respect to patients younger than 60 years (reference group). Tumor size ($p = 0.1990$) and TIL grade ($p = 0.2448$) were not associated with OS in the study cohort. Secondary models investigating the effects of RT that targeted draining lymph nodes compared with RT that did not demonstrated no significant association between RT that targeted draining lymph nodes and OS in the study cohort ($p = 0.8614$).

Immune Status and Other Clinical Factors

Immune status was significantly associated with odds of multiple pathologic factors including LVI, number of positive lymph nodes, and ECS. Immunosuppressed patients in our cohort were at increased odds of having LVI (odds ratio [OR], 1.58; 95% 1.23-2–03), and had increased odds of having involved lymph nodes (proportional OR, 1.70; 95% CI, 1.37–2.11, associated with increased odds of moving to a higher nodal category, i.e., from no positive nodes to 1 to 3 positive lymph nodes). Immunosuppressed patients were at decreased odds of having ECS (OR, 0.70; 95% CI, 0.50–0.99). Immune status was not associated with odds of greater tumor size or TIL grade ($p > 0.05$). Rates of radiation, chemotherapy, and nodal RT were also similar between IS and IC when control was used for other demographic and clinical factors ($p > 0.05$).

DISCUSSION

The association between immunosuppression, increased incidence of MCC, and worsened clinical outcomes for IS patients has been characterized in multiple studies.[5,7,21–25] Recommendations from the AJCC acknowledge these associations and encourage practitioners to consider immune status in both treatment and surveillance of patients with MCC, concordant with national guidelines (NCCN). However, immunosuppression is not incorporated into current AJCC staging, with AJCC authors acknowledging the subjective nature of immunosuppression as a prognostic factor limiting inclusion.

In this study, we performed a national database analysis to further define immune status as a prognostic factor in MCC, which to our knowledge represents the largest cohort of patients with localized MCC and known immune status to date. We observed immune status to be an independent predictor of OS in our cohort with adjustment for a priori selected clinically relevant factors. Secondary models suggested that patients with a solid-organ transplant may have worse survival than those with other immunosuppressive etiologies including CLL, NHL, and HIV/AIDS.

A differential effect of immune status on OS by age was appreciated in the cohort. Immunosuppression also was associated with increased likelihood of multiple other adverse clinical factors such as greater positive lymph node burden and lymphovascular invasion.

The results of this study demonstrated worsened 3-year OS rates and increased mortality hazard with immunosuppression, which are concordant with prior reports.[5–7,25] Paulson et al.[5] performed a retrospective multi-institutional analysis of 471 patients (41 patients with immunosuppression) and showed that immunosuppression was associated with increased risk of mortality and worsened 3-year OS compared with immunocompetence. Tarantola et al.[7] performed a retrospective analysis of 240 patients with MCC (33 patients with known immunosuppression) and found that the immunosuppressed patients in their cohort had a significantly worse 3-year OS than the immunocompetent patients. In contrast, Hui et al.[26] did not find immunosuppression to be associated with worsened locoregional recurrence-free survival, progression-free survival, or overall survival in a retrospective analysis of 176 patients (14 patients with known immunosuppression).

The cohort in our study included more than 3000 patients (412 patients with known immunosuppression). The results of our study offer strong support for consideration of immune status as an important prognostic factor for the OS of patients with localized MCC. Regarding the noted differential effect of immune status on OS in our cohort by patient age, aging has been associated with complex functional and architectural changes in the immune system (i.e., immunosenescence) even when another etiology of profound immunosuppression is not present.[27] These changes in aggregate diminish response to immunologic challenges, impair development of long-term immunologic memory, and may at least contribute in part to the decreased effect of immune status on OS for patients 70 years of age or older compared with the younger patients in the study cohort.

Limited by relatively small numbers of immunosuppressed patients, multiple prior studies, including those discussed earlier, did not present analyses regarding etiology of immunosuppression and potential differential association with OS. Cook et al.[28] performed a retrospective analysis of 89 patients who had non-metastatic MCC with known immunosuppression (28 patients with CLL, 19 patients with a solid-organ transplant, 19 patients with autoimmune disease, 16 patients with other hematologic malignancies, and 7 patients with HIV/AIDS). In their series, etiology of immunosuppression was associated with OS, showing worsened Merkel cell carcinoma-specific survival for patients with HIV/AIDS or a solid-organ transplant relative to patients with other immunosuppressive etiologies. Similarly, etiology of immunosuppression was associated with OS in our study, with a worsened 3-year OS for the patients who had a solid-organ transplant relative to the patients with CLL, NHL or “other” (including HIV/AIDS) for etiology of immunosuppression.

Although immunosuppression has been associated with greater risk of recurrence and decreased survival for patients with MCC,[5,29] relationships between immune status and other known clinically relevant baseline tumor and patient factors have not been fully characterized in the available literature. Tseng et al.[30] noted the distribution of baseline patient demographic, pathologic, and treatment characteristics to be similar between immunosuppressed and immunocompetent patients in a retrospective analysis of 805 patients with non-metastatic MCC, but did not present analyses comparing the relationship between immune status and other clinical factors.

In the current study, immunosuppression was associated with increased likelihood of multiple adverse clinical factors including greater burden of positive lymph nodes. Conic et al.[31] noted numerically higher rates of sentinel lymph node (SLN) positivity for immunosuppressed patients with MCC relative to those with immunocompetence, although this difference was not statistically significant in their analysis, potentially due to sample size limitations (71 patients with IS undergoing SLN examination). Similarly, immunosuppression has been associated with greater risk of metastasis for patients with cutaneous squamous cell carcinoma.[32]

Immunosuppression also was associated with greater likelihood of LVI in our study cohort. Although LVI has been associated with worsened OS[33,34] and greater risk of recurrence, to our knowledge the relationship between immune status and LVI has not been previously characterized for patients with MCC. Further studies are necessary to provide additional nuance regarding systemic immunosuppression, the tumor microenvironment, and LVI.[35,36]

STUDY LIMITATIONS

This study had specific limitations meriting further discussion besides those inherent to retrospective analysis in general. Immune status was not available for most patients in the overall dataset (pre-selection criteria) (Fig. S1), potentially limiting the generalizability of the study results to the overall population of patients with MCC in the United States. We were unable to characterize immunosuppression severity by quantification of circulating lymphocyte populations. Furthermore, more granular characterization of immunosuppressive etiology was unavailable in the dataset for the subgroup of IS patients coded as “other” including HIV/AIDS patients (102 patients with profound IS having an NOS etiology, 9 patients with HIV/AIDS, and 5 patients with profound IS from multiple conditions). The results of our study regarding IS may not be applicable to patients with MCC and well-controlled HIV. Information regarding the cellular composition of tumor infiltrates using immunohistochemical staining was not available in the dataset. Subsequently, we were unable to characterize the cellular composition of tumor infiltrates in our cohort or to determine the association between individual cell types within the infiltrate and OS.

We were unable to adjust for inter-reviewer variability in TIL grade-scoring potentially present in the dataset, although high rates of interobserver agreement regarding TIL grade have previously been reported.[37] Merkel cell polyomavirus positivity was unavailable in the dataset and could not be adjusted for in the performed analyses. Although we attempted to characterize the association of RT that targeted draining lymph nodes and OS in the study cohort, granular details regarding radiation field design were unavailable, limiting more robust analysis of optimal radiation target volumes. For this question, further locoregional control is a more optimal end point to evaluate than survival. The time frame for our study cohort predated the initial Food and Drug Administration (FDA) approval for immunotherapy for patients with metastatic MCC in 2017. Future studies are necessary to characterize outcomes in the advent of more routinely used immunotherapy for patients with locally advanced or metastatic MCC.

The NCDB collects treatment details regarding the initial course of therapy, and we were unable to adjust for salvage therapies potentially modulating overall survival. Cancer-specific clinical outcomes of interest including MCC-specific-survival, progression-free survival, and locoregional control are unavailable in the dataset and could not be examined. Both IS and IC patients with MCC may have competing risks for mortality, such as intercurrent illness, limiting OS as a clinical end point.[38] Despite this acknowledged limitation, OS still may be a clinical end point of interest for patients with MCC, particularly for IS patients, with Tseng et al.[30] demonstrating a strong correlation between OS and MCC-specific survival for IS patients in their cohort (79% of deaths attributable to MCC for IS patients).

CONCLUSIONS

We performed an analysis of a national database including, to our knowledge, the largest cohort of patients with localized MCC and known immune status and found immune status to be an independent predictor of OS in our cohort. Etiology of immunosuppression also was associated with differential effects on OS in our cohort. The results of this study in aggregate support immune status as an important predictor of OS for patients with localized MCC and consideration for inclusion of immune status in future MCC staging classifications. Additional studies are needed for quantitative characterization of immunosuppression and clinical outcomes in MCC, as well as for validation of the associations between immune status and the aforementioned clinical factors demonstrated in this study cohort. Further studies including potential multi-institutional analyses with granularity to characterize immune status as a predictor of cancer-specific clinical outcomes also are imperative.

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REFERENCES

1. Paulson KG, Park SY, Vandeven NA, et al. Merkel cell carcinoma: current U.S. incidence and projected increases based on changing demographics. *J Am Acad Dermatol.* 2018;78:457–463.e452.
2. Ma JE, Brewer JD. Merkel cell carcinoma in immunosuppressed patients. *Cancers Basel*. 2014;6:1328–50.
3. Clarke CA, Robbins HA, Tatalovich Z, et al. Risk of Merkel cell carcinoma after solid organ transplantation. *J Natl Cancer Inst*. 2015;107(2):1–9.
4. Bhatia S, Afanasiev O, Nghiem P. Immunobiology of Merkel cell carcinoma: implications for immunotherapy of a polyomavirus-associated cancer. *Curr Oncol Rep*. 2011;13:488–97.
5. Paulson KG, Iyer JG, Blom A, et al. Systemic immune suppression predicts diminished Merkel cell carcinoma-specific survival independent of stage. *J Invest Dermatol*. 2013;133:642–6.
6. Asgari MM, Sokil MM, Warton EM, Iyer J, Paulson KG, Nghiem P. Effect of host, tumor, diagnostic, and treatment variables on outcomes in a large cohort with Merkel cell carcinoma. *JAMA Dermatol*. 2014;150:716–23.
7. Tarantola TI, Vallow LA, Halyard MY, et al. Prognostic factors in Merkel cell carcinoma: analysis of 240 cases. *J Am Acad Dermatol*. 2013;68:425–32.
8. Bichakjian CK, Olencki T, Aasi SZ, et al. Merkel Cell Carcinoma, version 1.2018, NCCN Clinical Practice Guidelines in Oncology. *J Natl Compr Cancer Netw*. 2018;16:742–74.
9. Trinidad CM, Torres-Cabala CA, Prieto VG, Aung PP. Update on eighth-edition American Joint Committee on Cancer classification for Merkel cell carcinoma and histopathological parameters that determine prognosis. *J Clin Pathol.* 2019;72:337–40.
10. Bhatia S, Storer BE, Iyer JG, et al. Adjuvant radiation therapy and chemotherapy in Merkel cell carcinoma: survival analyses of 6908 cases from the National Cancer Database. *J Natl Cancer Inst.* 2016;108(9):1–9.
11. Bilimoria KY, Stewart AK, Winchester DP, Ko CY. The National Cancer Database: a powerful initiative to improve cancer care in the United States. *Ann Surg Oncol*. 2008;15:683–90.
12. Ruhl J, Hofferkamp J, Ward E, et al. Site-Specific Data Item (SSDI) manual. Springfield, IL: North American Association of Central Cancer Registries; 2019.
13. Agarwal P, Moshier E, Ru M, et al. Immortal time bias in observational studies of time-to-event outcomes: assessing effects of postmastectomy radiation therapy using the National Cancer Database. *Cancer Control*. 2018;25:1073274818789355.
14. Park HS, Gross CP, Makarov DV, Yu JB. Immortal time bias: a frequently unrecognized threat to validity in the evaluation of postoperative radiotherapy. *Int J Radiat Oncol Biol Phys*. 2012;83:1365–73.
15. Grambsch PM, Therneau TM. Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*. 1994;81:515–26.
16. Eisemann N, Waldmann A, Katalinic A. Imputation of missing values of tumour stage in population-based cancer registration. *BMC Med Res Methodol*. 2011;11:129.
17. van Buuren S, Groothuis-Oudshoorn K. Mice: multivariate imputation by chained equations in R. *J Stat Softw.* 2011;45(3):1–67.
18. Robitzsch A, Grund S. miceadds: Some Additional Multiple Imputation Functions, Especially for 'mice'. R package version 3.11-6. 2021. https://CRAN.R-project.org/package=miceadds.
19. R Core Team. R: a language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing; 2017. https://www.R-project.org/.
20. von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *Int J Surg*. 2014;12:1495–9.
21. Penn I, First MR. Merkel's cell carcinoma in organ recipients: report of 41 cases. *Transplantation*. 1999;68:1717–21.
22. An KP, Ratner D. Merkel cell carcinoma in the setting of HIV infection. *J Am Acad Dermatol*. 2001;45:309–12.
23. Brewer JD, Shanafelt TD, Otley CC, et al. Chronic lymphocytic leukemia is associated with decreased survival of patients with malignant melanoma and Merkel cell carcinoma in a SEER population-based study. *J Clin Oncol*. 2012;30:843–9.
24. Garrett GL, Blanc PD, Boscardin J, et al. Incidence of and risk factors for skin cancer in organ transplant recipients in the United States. *JAMA Dermatol*. 2017;153:296–303.
25. Liang E, Brower JV, Rice SR, Buehler DG, Saha S, Kimple RJ. Merkel cell carcinoma analysis of outcomes: a 30-year experience. *PloS One*. 2015;10:e0129476.
26. Hui AC, Stillie AL, Seel M, Ainslie J. Merkel cell carcinoma: 27-year experience at the Peter MacCallum Cancer Centre. *Int J Radiat Oncol Biol Phys*. 2011;80:1430–5.
27. Ladomersky E, Zhai L, Lauing KL, et al. Advanced age increases immunosuppression in the brain and decreases immunotherapeutic efficacy in subjects with glioblastoma. *Clin Cancer Res*. 2020;26:5232–45.
28. Cook M, Baker K, Redman M, et al. Differential outcomes among immunosuppressed patients with Merkel cell carcinoma: impact of immunosuppression type on cancer-specific and overall survival. *Am J Clin Oncol*. 2019;42:82–8.
29. Bryant MK, Ward C, Gaber CE, Strassle PD, Ollila DW, Laks S. Decreased survival and increased recurrence in Merkel cell carcinoma significantly linked with immunosuppression. *J Surg Oncol*. 2020;122:653–9.
30. Tseng YD, Nguyen MH, Baker K, et al. Effect of patient immune status on the efficacy of radiation therapy and recurrence-free

survival among 805 patients with Merkel cell carcinoma. *Int J Radiat Oncol Biol Phys*. 2018;102:330–9.
31. Conic RRZ, Ko J, Saridakis S, et al. Sentinel lymph node biopsy in Merkel cell carcinoma: predictors of sentinel lymph node positivity and association with overall survival. *J Am Acad Dermatol*. 2019;81:364–72.
32. Thompson AK, Kelley BF, Prokop LJ, Murad MH, Baum CL. Risk factors for cutaneous squamous cell carcinoma recurrence, metastasis, and disease-specific death: a systematic review and meta-analysis. *JAMA Dermatol*. 2016;152:419–28.
33. Fields RC, Busam KJ, Chou JF, et al. Recurrence and survival in patients undergoing sentinel lymph node biopsy for Merkel cell carcinoma: analysis of 153 patients from a single institution. *Ann Surg Oncol*. 2011;18:2529–37.
34. Harounian JA, Molin N, Galloway TJ, et al. Effect of sentinel lymph node biopsy and LVI on Merkel cell carcinoma prognosis and treatment. *Laryngoscope*. 2020;131(3):E828–35.
35. Kirita K, Ishii G, Matsuwaki R, et al. Identification of biological properties of intralymphatic tumor related to the development of lymph node metastasis in lung adenocarcinoma. *PloS One*. 2013;8:e83537.
36. Ikemura S, Aramaki N, Fujii S, et al. Changes in the tumor microenvironment during lymphatic metastasis of lung squamous cell carcinoma. *Cancer Sci*. 2017;108:136–42.
37. Paulson KG, Iyer JG, Tegeder AR, et al. Transcriptome-wide studies of Merkel cell carcinoma and validation of intratumoral CD8+ lymphocyte invasion as an independent predictor of survival. *J Clin Oncol*. 2011;29:1539–46.
38. Fields RC, Busam KJ, Chou JF, et al. Five hundred patients with Merkel cell carcinoma evaluated at a single institution. *Ann Surg*. 2011;254:465–73.